

Independent verification of dilogarithm ladders and Sun's conjecture (arXiv:2506.10206)

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Abstract

We independently re-verify, to ≥ 38 digits, the principal dilogarithm identities of C. Hakimoglu-Brown, *Analytic Dilogarithm Identities*, arXiv:2506.10206 (June 2025): the $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{2})$ two-term Li_2 -relations, the famous Lima and Zagier ladder identities, and—most notably—the resolution of an *open conjecture of Z.-W. Sun*. The paper's engine, the D'Aurizio-style inverse-binomial ${}_4F_3$ -series, is the natural sibling of the central-binomial ladder of `harvested-central-binomial-arcsin.wl`.

1 What was verified

All to ≥ 38 digits (`tools/scratch/dilog-inverse-binomial/`); a representative selection:

- **Sun's open conjecture** (eq. (24)), proved in the paper via the $\mathbb{Q}(\sqrt{3})$ evaluation (25):

$$\sum_{n \geq 0} \frac{\binom{2n}{n}}{(2n+1)8^n} \left(\sum_{0 \leq k < n} \frac{(-1)^k}{2k+1} - \frac{(-1)^n}{2n+1} \right) = -\frac{\sqrt{2}}{16} \pi^2.$$

- **Theorem 2** ($\mathbb{Q}(\sqrt{3})$): $8 \text{Li}_2(2 - \sqrt{3}) - \text{Li}_2(4\sqrt{3} - 7) = \frac{5\pi^2}{12} + \ln(2 - \sqrt{3}) \ln(2 + \sqrt{3})$, and $3 \text{Li}_2(7 - 4\sqrt{3}) = 6 \text{Li}_2(2 - \sqrt{3}) + 6 \text{Li}_2(\sqrt{3} - 2)$.
- **Lima's identity** ($\mathbb{Q}(\sqrt{2})$, eq. (26)): $\text{Li}_2(\frac{1}{\sqrt{2}}) - \text{Li}_2(1 - \sqrt{2}) = \frac{\pi^2}{6} - \frac{\ln^2 2}{8} + \frac{1}{2} \ln 2 \ln(\sqrt{2} - 1)$.
- **Zagier's four-term** (eq. (27)): $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{\pi^2}{12}$, with the (unnormalized) Rogers $L(x) = \text{Li}_2(x) + \frac{1}{2} \ln |x| \ln |1 - x|$; and its compressed three-term form.
- the rational-argument $\frac{\pi^2}{18} + 2 \ln 2 \ln 3 - 2 \ln^2 2 - \frac{2}{3} \ln^2 3 = \text{Li}_2(\frac{1}{4}) + \frac{1}{3} \text{Li}_2(\frac{1}{9})$, the inverse-tangent/ $\text{Cl}_2(\frac{\pi}{3})$ relation (30), and $\text{Li}_2(\frac{4}{3}) + \text{Li}_2(4) = \frac{1}{2}(\pi^2 - \ln^2 3)$ (real parts).
- the **engine**: ${}_4F_3(\{1, 1, 1, \frac{3}{2}\}; \{\frac{5}{2}, \frac{5}{2}, \frac{5}{2}\}; 1) = \frac{27}{2}(7\zeta(3) + (3 - 2G)\pi - 12)$ (D'Aurizio) and the cube-binomial ${}_4F_3$ of eq. (1).
- the paper's **claimed-original elliptic** ${}_4F_3(1)$ **evaluations** (eqs. 32, 33, 35):

$$\begin{aligned} \text{Cl}_2\left(\frac{\pi}{3}\right) &= \frac{\sqrt{3}}{2} {}_4F_3\left(\left\{1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right\}; \left\{\frac{3}{2}, \frac{5}{6}, \frac{7}{6}\right\}; 1\right), \\ {}_4F_3\left(\left\{1, \frac{1}{2}, \frac{2}{3}, \frac{1}{3}\right\}; \left\{\frac{3}{2}, \frac{5}{4}, \frac{3}{4}\right\}; 1\right) &= \frac{3\sqrt{3}}{4} (2 \ln(4 + 3\sqrt{2}) + \ln(\frac{7}{2} - 2\sqrt{3})), \\ \ln^2\left(\frac{5+\sqrt{13}}{2\sqrt{3}}\right) - \frac{\pi^2}{12} &= \frac{1}{180} {}_4F_3\left(\left\{1, 1, 1, \frac{3}{2}\right\}; \left\{\frac{7}{6}, \frac{11}{6}, 2\right\}; -\frac{1}{324}\right). \end{aligned}$$

All check out. One caution worth recording: equation (27) uses the *unnormalized* Rogers L (so the right side is $-\frac{\pi^2}{12}$, not a pure rational); with the $\frac{6}{\pi^2}$ -normalized L the identity reads $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{1}{2}$.

Connection. The paper evaluates inverse-binomial series with $\binom{2n}{n}^2$ and $\binom{3n}{n}$ via beta integrals; this is the reciprocal-binomial sibling of the central-binomial ladder $\sum \binom{2n}{n} 4^{-n} (2n+1)^{-p}$ studied in `harvested-central-binomial-arcsin.wl`. The verified identities are stored (with attribution) in `src/PolyLog/identities/harvested-dilogarithm-ladders-2506.10206.wl`.